Dual-Feature Attention Hybrid GCN Mamba Network for Joint Hyperspectral and LiDAR Classification

Zhenyang Xie, Li Lv, Hongmin Gao, *Senior Member, IEEE*, Shufang Xu, *Member, IEEE*, and Haihua Xie

***Abstract*—Hyperspectral images (HSIs) and light detection and ranging (LiDAR) data provide complementary spectral–spatial and elevation information, respectively, whose fusion can significantly improve classification accuracy. However, their inherent heterogeneity challenges effective spectral–geospatial integration. Although convolutional neural networks (CNNs), recurrent neural networks (RNNs), and transformer models have advanced multimodal remote sensing classification, each shows limitations. CNNs excel in spatial aggregation but lack global context, while RNNs struggle with long-range dependencies, and transformers face computational inefficiency despite capturing long-range spectral features. To address these limitations, we propose a dual-feature attention hybrid graph convolutional network (GCN) Mamba network (DAHGMN) for joint HSI and LiDAR classification. Specifically, multimodal image cubes are first extracted by a CNN to obtain initial features. Subsequently, a dual-feature attention (DA) module is introduced to adaptively recalibrate spectral and spatial feature weights, enhancing discriminability. Furthermore, we propose a hybrid GCN Mamba (HGM) module with both low parameter complexity and time complexity, which combines the local geometric modeling capability of GCNs with the global long-range dependency modeling of Mamba's state-space model (SSM). A probability-based decision fusion strategy is employed to integrate multilevel classification results, achieving an efficient combination of the spatial–spectral contextual features. Extensive experiments on three benchmark HSI-LiDAR datasets demonstrate that DAHGMN achieves superior classification accuracy while significantly reducing parameter complexity compared to state-of-the-art methods. The implementation code is publicly available at https://github.com/RogsXie/DAHGMN**

***Index Terms*—Attention mechanism, graph convolution network (GCN), joint classification, light detection and ranging (LiDAR), Mamba.**

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Zhenyang Xie is with the School of Information Engineering, Jiangxi University of Water Resources and Electric Power, Nanchang 330099, China (e-mail: Xiewak@163.com).

Li Lv and Haihua Xie are with the School of Information Engineering, Jiangxi University of Water Resources and Electric Power, Nanchang 330099, China, and also with Jiangxi Province Engineering Research Center for Intelligent Processing and Early Warning Technology of Water Conservancy Big Data, Nanchang 330099, China (e-mail: lvli@nit.edu.cn; xie_haihua@nit.edu.cn).

Hongmin Gao is with the College of Computer Science and Software Engineering, Hohai University, Nanjing 211100, China (e-mail: gaohongmin@hhu.edu.cn).

Shufang Xu is with the College of Computer Science and Software Engineering, Hohai University, Nanjing 211100, China, and also with Shaanxi Key Laboratory of Optical Remote Sensing and Intelligent Information Processing, Xi'an 710119, China (e-mail: xushufang@hhu.edu.cn).

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I. Introduction

HYPERSPECTRAL images (HSIs), as high-dimensional imaging data, capture both spatial distribution characteristics and fine spectral information of ground objects through continuous narrow-band imaging [1], [2], and have been widely applied in land cover change detection [3], vegetation monitoring [4], ecological assessment [5], resource management [6], and geological exploration [7]. However, HSI-based land cover classification still faces challenges including spectral mixing and limited spatial resolution. The rapid development of remote sensing acquisition systems has led to substantial growth in the accessibility of multisensor Earth observation datasets, where different sensors capture complementary information: HSIs provide detailed spectral characteristics [8], light detection and ranging (LiDAR) delivers precise terrain elevation [9], and synthetic aperture radar (SAR) acquires geometric amplitude and phase information [10]. Research demonstrates that multisource data fusion enhances information completeness and classification accuracy [11], with LiDAR being particularly valuable for its ability to accurately measure 3-D structural features and maintain robust elevation data acquisition in complex environments [12]. Consequently, the effective extraction and fusion of spectral–spatial features from HSI-LiDAR data for classification performance enhancement [13], [14] has emerged as a critical research focus in remote sensing.

In supervised classification research, the evolution of HSI classification methods over the past decade has progressed from traditional machine learning to deep learning (DL) approaches to overcome spectral information limitations. Traditional methods, such as random forests (RFs) [15], k-nearest neighbors (KNNs) [16], and support vector machines (SVMs) [17], typically treat HSI pixels as independent spectral signatures and utilize raw spectral bands or manually designed spectral descriptors, ignoring spatial context information, thus limiting the classification accuracy. Furthermore, these methods often demonstrate limited robustness against noise interference and missing data. The rapid advancement of DL

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has introduced powerful nonlinear representation capabilities, enabling automatic extraction of high-level semantic features and demonstrating superior performance in HSI classification. Various DL architectures have been successfully applied, including autoencoders (AEs) [18], convolutional neural networks (CNNs) [19], [20], [21], generative adversarial networks (GANs) [22], [23], recurrent neural networks (RNNs) [24], capsule networks (CapsNets) [25], graph convolutional networks (GCNs) [26], [27], and transformers [28]. Despite these advancements, current methods still face significant challenges in complex land cover classification scenarios, particularly in urban environments where spectrally similar construction materials in roads and buildings create substantial discrimination difficulties for classifiers.

In the joint classification of HSI and LiDAR data, DL frameworks have demonstrated a distinct evolutionary progression, with CNNs emerging as the predominant approach due to their exceptional feature extraction capabilities. The pioneering work in [29] first introduced extended morphological profiles of HSI and LiDAR data into CNNs for deep semantic feature extraction, establishing a DL classification paradigm. Subsequent advancements include the dual-branch CNN architecture proposed by Xu et al. [30], which processes modalities separately before feature concatenation, and the coupled CNN framework developed by Hang et al. [31] that employs weight-shared branches for collaborative feature extraction with both feature-level and decision-level fusion strategies. Despite achieving competent classification, this method produces spatially oversmoothed results due to limited feature discriminability. MSNetSC [32] addresses this through multiscale self-calibrated convolutions with attention-based contextual fusion. Subsequently, Wu et al. [33] developed a CNN framework with cross-channel reconstruction for enhanced multimodal remote sensing data fusion. Zhang et al. [34] further proposed an interleaved perception CNN to integrate heterogeneous information for improved classification, while Gao et al. [35] proposed an adaptive multiscale spatial–spectral enhancement network (AMSSE-Net). Although these CNN-based fusion methods demonstrate remarkable capabilities in automatically learning representative features and have significantly advanced multimodal classification, they remain fundamentally limited by their focus on local spatial features, failing to adequately capture the global sequential characteristics inherent in spectral features. This constraint ultimately limits their classification performance potential.

Compared with CNNs, recurrent architectures exhibit enhanced capability in analyzing spectral sequences by leveraging the fine spectral resolution characteristics of HSI to model long-distance feature relationships. Zhou et al. [36] pioneered the use of cascaded RNN architectures to eliminate interband spectral redundancy while comprehensively extracting complementary features. More recently, the emergence of transformer architectures has revolutionized HSI classification through their exceptional global modeling capabilities, beginning with the groundbreaking application of vision transformers (ViTs) [37]. Subsequent developments include Yang et al. [38] hybrid GCN-transformer network that enhances discrimination of subtle spectral variations, and Xie et al. [39] semantic and spectral–spatial feature fusion transformer (S3FFT) enabling end-to-end integration of low-level spectral–spatial features with high-level semantics. Meanwhile, these recent advances have also driven progress in HSI and LiDAR joint classification. Roy et al. [40] multimodal fusion transformer (MFT) overcomes traditional transformers' limitations through its multihead cross-patch attention mechanism, which effectively integrates complementary data to produce more discriminative classification tokens. Zhao et al. [41] hierarchical CNN-transformer (HCT) addresses heterogeneous data fusion via a parallel architecture combining CNN-based local feature extraction with transformer-based global modeling. The global–local transformer network (GLT-Net) [42] further enhances this through a unified framework capturing multiscale relationships via convolutional local learning and transformer-based global analysis. Collectively, these advancements have significantly propelled the development of HSI classification methodologies while establishing novel paradigms for multimodal remote sensing data fusion research.

However, existing network architectures confront three fundamental limitations.

1) Some current methodologies still rely on shallow CNN architectures that extract only elementary features from HSI and LiDAR data, subsequently implementing basic feature fusion strategies.

2) Despite transformers' demonstrated efficacy in long-range dependency modeling and multimodal representation enhancement, their quadratic computational complexity scaling with input sequence length imposes severe efficiency constraints.

3) The method of modern network design using serial connection easily leads to insufficient interaction between local and global features, which hinders effective attention to the spectrally discriminative regions.

Recent advances in state-space models (SSMs) [43] have led to the development of the Mamba framework [44], which achieves an optimal balance between global information modeling and computational efficiency while demonstrating state-of-the-art performance across multiple benchmarks [45]. This SSM-based approach fundamentally addresses the computational efficiency limitations inherent in both RNNs and transformers, offering novel perspectives for sequence modeling. The architecture's key advantage stems from its linear scaling of computational complexity and memory requirements with sequence length, yielding significant improvements in both training and inference efficiency. Gu et al. [46] further enhanced long-range dependency modeling through the high-order polynomial projection operators (HiPPO) mechanism, which compresses input signals into compact coefficient vectors while optimizing memory update strategies for efficient sequence processing. Subsequent improvements in the structured state-space sequence model (S6) simplified parameterization methods, enabling streamlined inference for ultralong sequences. The Mamba architecture exhibits enhanced computational efficiency relative to conventional transformer models of comparable scale in linguistic and

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acoustic applications, attributable to its selective feature integration mechanism, streamlined network design, and efficient scanning methodology. The success of Mamba in sequential modeling has motivated its adaptation to remote sensing applications. For HSI classification, HyperMamba [47] introduces spectrally–spatially adaptive scanning through two innovative modules, effectively addressing transformers' limitations in high-dimensional data processing. In scene classification tasks, RSMamba [48] resolves the noncausal 2-D data challenge via a dynamic multipath activation mechanism, demonstrating Mamba's capability in modeling complex spatial relationships. Extending Mamba to multimodal fusion, HLMamba [49] integrates HSI and LiDAR data through a gradient joint algorithm and multimodal Mamba fusion module, significantly enhancing feature representation by leveraging cross-modal dependencies. Meanwhile, S2CrossMamba [50] proposes an inverted bottleneck cross-SSM architecture to dynamically fuse multimodal features with centralized state updates. Collectively, these works underscore Mamba's potential as a computationally efficient alternative to transformers in remote sensing applications. However, implementing SSMs for HSI analysis presents notable difficulties stemming from the spectral dimensionality and intrinsic data redundancy inherent to HSI.

The application of GCNs [51] in HSI classification has shown that compared with traditional CNNs, they exhibit greater flexibility in capturing unstructured spatial features. particularly in capturing unstructured spatial features through their ability to directly process non-Euclidean geometric structures and explicitly construct adjacency matrices for pixel-level feature aggregation [52]. To enhance global context understanding, Ding et al. [53] developed a GraphSAGE-based multiscale graph aggregation network that autonomously learns deep semantic representations from HSIs. Recent hybrid approaches combining CNNs and GCNs have emerged to address CNN's small-sample limitations and GCN's computational intensity, exemplified by the complementary edge-guided GCN (CEGCN) [54] that jointly extracts pixel-level and superpixel-level spectral–spatial features. However, despite these advancements, current GCN architectures remain fundamentally constrained by their adjacency matrices' exclusive focus on spatial relationships, which neglects spectral correlation constraints and consequently limits the full exploitation of HSI data's inherent joint spectral–spatial characteristics.

To address these challenges while achieving adaptive mining of spatial–spectral feature relationships and efficient parallel computation, we explore the potential of integrating the Mamba architecture with GCNs for multimodal remote sensing classification. We propose a novel dual-feature attention hybrid GCN Mamba network (DAHGMN) for joint HSI and LiDAR classification. Specifically, DAHGMN adopts a CNN-based feature extractor to extract fine features and introduces dual-feature attention (DA) module to enhance discriminative feature representation. Simultaneously, a low-parameter, low-complexity hybrid GCN Mamba (HGM) module is designed to combine GCN's local spatial pattern modeling with Mamba's global dependency capture. Rigorous evaluation on three benchmark HSI-LiDAR datasets confirms the consistent superiority of our proposed framework over existing approaches, as validated through both quantitative metrics and visual assessments.

The principal contributions of this work encompass.

1) We design an integrated feature extraction framework incorporating principal component analysis (PCA) and 3-D convolution, and introduce a DA module for feature refinement. This architecture enables comprehensive spectral–spatial feature extraction and fusion through multiperspective analysis and dimensionality reduction.

2) We propose an HGM module that integrates SpectralSSM within the Mamba block. The SpectralSSM employs multiscale adaptive time steps to address spectral dimensionality challenges, while triple-path residuals and selective state updates reduce data redundancy. The GCN block utilizes dynamic adjacency matrices, eliminating reliance on static structures and combining with the Mamba branch to compensate for the overlooked interspectral correlation constraints. This module achieves significantly reduced parameter complexity and computational requirements while effectively combining global spectral modeling with local feature extraction.

3) We validate DAHGMN's superior performance through extensive experiments on three benchmark datasets (Houston, MUUFL, and Trento), demonstrating consistent improvements over state-of-the-art methods, with ablation studies quantitatively verifying the contribution of each key component.

The article is structured as follows. Section II establishes the theoretical framework, detailing the fundamental principles of Mamba's SSMs and GCNs; Section III presents the proposed DAHGMN framework, systematically describing its architectural design and core components; Section IV conducts comprehensive experimental validation, including ablation studies, quantitative performance comparisons, and qualitative visual analyses across multiple datasets; finally, Section V concludes with key findings and proposes promising directions for future research in multimodal remote sensing classification.

II. Review on SSM and GCN

A. State-Space Models

SSMs represent fundamental dynamical systems in control theory that employ intermediate state variables to establish linear time-invariant relationships between input $x(t) \in R$, state $h(t) \in R^N$, and output $y(t) \in R$ vectors. The structured state-space sequence model (S4), derived from SSM principles, implements nonlinear sequence-to-sequence mappings through continuous-time system representations

$$\begin{aligned} h'(t) &= \mathbf{A}h(t) + \mathbf{B}x(t) \\ y(t) &= \mathbf{C}h(t) \end{aligned} \tag{1}$$

where $\mathbf{A} \in R^{N\times N}$ denotes the state transition matrix, and $\mathbf{B} \in R^{N\times 1}$ and $\mathbf{C} \in R^{1\times N}$ represent projection matrices.

For digital image processing applications, the continuous-time SSM requires discretization to process discrete spatial sequences. This transformation is formally accomplished

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through zeroth-order hold (ZOH) discretization with sampling interval Δ, yielding the discrete-time system representation

$$h_k = \overline{\mathbf{A}} h_{k-1} + \overline{\mathbf{B}} x_k$$
$$y_k = \overline{\mathbf{C}} h_k \tag{2}$$

where the discretized parameters are derived as $\overline{\mathbf{A}} = e^{\Delta \mathbf{A}}$, $\overline{\mathbf{B}} = (\Delta \mathbf{A}^{-1}(e^{\Delta \mathbf{A}} - \mathbf{I}))\Delta \mathbf{B}$, and $\overline{\mathbf{C}} = \mathbf{C}$ maintaining their continuous-time forms, and where k denotes the discrete spatial index.

To optimize computational efficiency in large-scale remote sensing applications, SSMs leverage discrete convolution operations to accelerate the linear recurrence relations through the following transformation:

$$\overline{K} = \left(\overline{\mathbf{C}}\overline{\mathbf{B}}, \overline{\mathbf{C}}\overline{\mathbf{A}}\overline{\mathbf{B}}, \ldots, \overline{\mathbf{C}}\overline{\mathbf{A}}^{L-1}\overline{\mathbf{B}}\right)$$
$$y = x * \overline{K} \tag{3}$$

where L denotes the sequence length, * represents the discrete convolution operator, and $\overline{K} \in R^L$ constitutes the SSM kernel formed by the discretized system parameters $\overline{\mathbf{A}}$, $\overline{\mathbf{B}}$, and $\overline{\mathbf{C}}$, enabling efficient parallel computation through modern DL frameworks.

B. Graph Convolutional Network

GCNs explicitly model spatial relationships between different land cover types by representing data as undirected graphs. For a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$, where $\mathcal{V}$ denotes the vertex set with $|\mathcal{V}| = N$, $\mathcal{E}$ represents the edge set, and $\mathbf{A} \in R^{N \times N}$ is the adjacency matrix, the graph Laplacian $\mathbf{L} \in R^{N \times N}$ is defined as

$$\mathbf{L} = \mathbf{D} - \mathbf{A} \tag{4}$$

where $\mathbf{D} \in R^{N \times N}$ is the degree matrix of $\mathcal{G}$. The normalized Laplacian matrix is obtained by

$$\mathbf{L}_n = \mathbf{I} - \mathbf{D}^{-\frac{1}{2}} \mathbf{A} \mathbf{D}^{-\frac{1}{2}} \tag{5}$$

where $\mathbf{I} \in R^{N \times N}$ is the identity matrix.

GCN employ a convolutional kernel $\mathbf{g}_\theta = \mathrm{diag}(\theta)$ for node embedding, where $\theta \in R^N$. The graph convolution operation can be expressed as

$$\mathbf{x} * \mathbf{g}_\theta = \mathbf{U} \mathbf{g}_\theta (\mathbf{\Lambda}) \mathbf{U}^T \mathbf{x} \tag{6}$$

where $\mathbf{x}$ is the input signal, $\mathbf{U} \in R^{N \times N}$ contains the eigenvectors of $\mathbf{L}_n$, and $\mathbf{\Lambda}$ represents the corresponding eigenvalues. To simplify computation, Chebyshev polynomials approximate $\mathbf{g}_\theta$

$$\mathbf{g}_\theta (\mathbf{\Lambda}) \approx \sum_{k=0}^{K} \theta_k T_k (\mathbf{L}_n) \tag{7}$$

where T_k denotes the kth-order Chebyshev polynomial. Using first-order approximation yields

$$\mathbf{x} * \mathbf{g}_\theta = \theta \left(\widetilde{\mathbf{D}}^{-\frac{1}{2}} \widetilde{\mathbf{A}} \widetilde{\mathbf{D}}^{-\frac{1}{2}}\right) \mathbf{x}_{\mathcal{G}} \tag{8}$$

where $\widetilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ and $\widetilde{\mathbf{D}} = \sum_j \widetilde{\mathbf{A}}_{ij}$.

Based on (8), the layer-wise propagation rule for GCN is

$$\mathbf{H}^{(l+1)} = \sigma \left(\widetilde{\mathbf{D}}^{-\frac{1}{2}} \widetilde{\mathbf{A}} \widetilde{\mathbf{D}}^{-\frac{1}{2}} \mathbf{H}^{(l)} \mathbf{W}^{(l)}\right) \tag{9}$$

where $\mathbf{H}^{(l+1)} \in R^{N \times D'}$ and $\mathbf{H}^{(l)} \in R^{N \times D}$ represent the features at layers $l+1$ and l, respectively, $\sigma(\cdot)$ denotes the activation function, and $\mathbf{W} \in R^{D' \times D'}$ is the trainable weight matrix.

III. Methodology

This section elaborates on the comprehensive structural design of the DAHGMN framework, with its complete architecture visualized in Fig. 1. The complete workflow consists of four sequential stages: feature extraction, feature fusion, feature processing, and classification. Specifically, during the feature extraction stage, a CNN-based feature extractor processes both data modalities to derive spectral–spatial features from HSI and elevation characteristics from LiDAR data. The second stage employs a DA module to enhance fused feature discriminability through adaptive spectral–spatial weighting. Subsequently, the HGM module in the processing stage simultaneously extracts and integrates both localized spatial and global spectral characteristics. Finally, the classification stage employs a fully connected (FC) layer classifier with softmax activation to execute decision fusion, generating the final classification results.

A. Stage of Feature Extraction

Recent advances have established CNNs as the predominant architecture for multimodal remote sensing data analysis, demonstrating superior capability in extracting joint spectral–spatial–elevation features. As illustrated in Fig. 1, our dual-stream CNN processes HSI and LiDAR data through specialized feature extraction stream. The HSI input tensor $\mathbf{X_H}$ contains d_h spectral bands with $\lambda \times \lambda$ spatial patches, while the LiDAR input tensor $\mathbf{X_L}$ maintains equivalent spatial dimensions with d_l elevation channels. The HSI processing stream sequentially executes PCA for spectral dimensionality reduction, 3-D convolutional processing with $3 \times 3 \times 3$ kernels, batch normalization, and Gaussian error linear unit (GeLU) activation followed by 2-D convolution to generate 64-channel spatial features. Concurrently, the LiDAR processing stream employs two successive 3×3 2-D convolutional layers with intermediate batch normalization and GeLU activation, where convolution kernels are empirically determined as 32 and 64. This architecture ensures dimensional alignment between modalities, yielding spatially congruent 64-channel feature maps from both HSI and LiDAR inputs to enable effective multimodal fusion in subsequent processing stages.

B. Stage of Feature Fusion

Effective feature fusion plays a crucial role in improving classification performance for HSI and LiDAR data integration. To address this challenge, we introduce a DA module that refines fused features through coordinated spectral and spatial attention operations. This mechanism is applied after the initial fusion of HSI and LiDAR features to enhance discriminative representation and improve model robustness. Let $\mathbf{F_h}$ and $\mathbf{F_l}$ denote the extracted HSI and LiDAR features, respectively, from the previous feature extraction stage. The fused feature $\mathbf{f}$ is obtained through adaptive linear weighting

$$\mathbf{f} = \alpha \mathbf{F_h} + (1 - \alpha) \mathbf{F_l} \tag{10}$$

where $\alpha \in [0, 1]$ represents a learnable weighting coefficient that is automatically optimized during network training. Considering the significant modality gap between HSI and LiDAR

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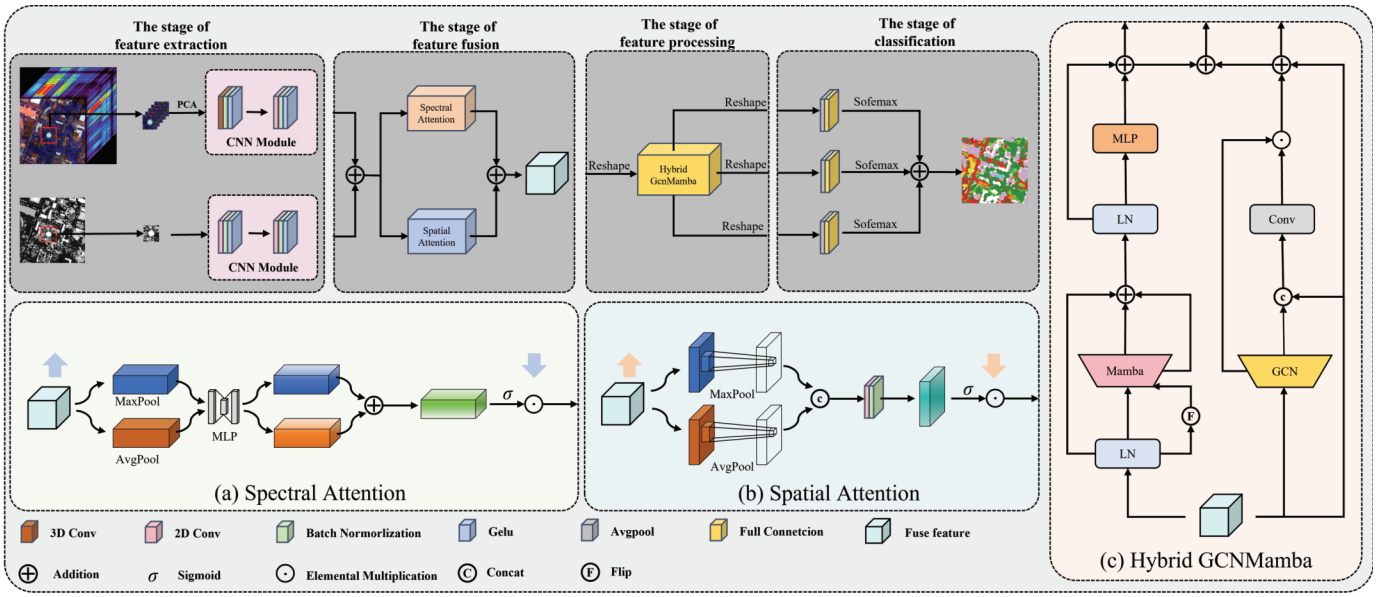


Fig. 1. Proposed framework of the DAHGMN for HSI and LiDAR data joint classification. (a) Spectral attention. (b) Spatial attention. (c) Hybrid GCNMamba.

data, we initialize α to 0.5 to ensure balanced contribution from both $\mathbf{F_h}$ and $\mathbf{F_l}$ in the early training phase. The DA module subsequently adjusts this balance based on the relative importance of spectral versus elevation information for each spatial location.

1) Spectral Attention: As illustrated in Fig. 1(a), the spectral attention module adaptively recalibrates feature responses by assigning distinct weights to each channel based on global contextual information. The module employs both global average pooling (GAP) and global max pooling (GMP) to capture complementary statistical descriptors of feature importance, followed by processing through a shared multilayer perceptron (MLP). Formally, given the initial fused feature map $\mathbf{f} \in R^{C\times H\times W}$ where C, H, and W denote channel count, height, and width, respectively, the spectral attention mechanism operates as

$$\mathbf{F}_{\text{Spe}} = \left(\sigma\left(\text{MLP}\left(\text{GAP}\left(\mathbf{f}\right)\right) + \text{MLP}\left(\text{GMP}\left(\mathbf{f}\right)\right)\right)\right)\mathbf{f} \tag{11}$$

where σ is the sigmoid activation function.

2) Spatial Attention: Fig. 1(b) illustrates the operation of the spatial attention mechanism, which improves discriminative feature extraction through selective region enhancement and background interference suppression. The module employs both channel-wise GAP and GMP to aggregate spatial information, concatenating their outputs to form a feature descriptor $\mathbf{M} \in R^{H\times W\times 2}$. This descriptor is then processed through a convolutional layer followed by sigmoid activation to generate spatial attention weights, which are subsequently applied to the input features $\mathbf{f}$. The complete process is formally expressed as

$$\begin{aligned}\mathbf{M} &= \text{Concat}\left(\text{GAP}\left(\mathbf{f}\right), \text{GMP}\left(\mathbf{f}\right)\right)\\ \mathbf{F}_{\text{Spa}} &= \left(\sigma\left(\text{Conv2D}_{7\times 7}\left(\mathbf{M}\right)\right)\right)\mathbf{f}\end{aligned} \tag{12}$$

where Concat$(\cdot)$ denotes channel-wise feature integration.

The final fused feature $\mathbf{F}_{\text{Fused}} \in R^{C\times H\times W}$ is obtained by integrating both spectral and spatial attention-enhanced features

$$\mathbf{F}_{\text{Fused}} = \mathbf{F}_{\text{Spe}} + \mathbf{F}_{\text{Spa}}. \tag{13}$$

This hierarchical attention mechanism enables simultaneous enhancement along both spectral and spatial dimensions, where $\mathbf{F}_{\text{Spe}}$ emphasizes discriminative spectral and $\mathbf{F}_{\text{Spa}}$ highlights spatially significant regions.

C. Stage of Feature Processing

As illustrated in Fig. 1(c), we propose an HGM module to process and combine heterogeneous features. This module employs a dual-branch hierarchical architecture that fully exploits spatial–spectral contextual information to enhance representation learning. The HGM consists of two parallel branches: a Mamba branch and a GCN branch.

1) Mamba Branch: To address the issues of long-distance spectral dependency and data redundancy, we proposed SpectralSSM. In SpectralSSM, we introduced multiscale adaptive step sizes. By dynamically integrating the step size parameters from different time scales, it achieves the ability to simultaneously capture high-frequency details and low-frequency trends. This mechanism automatically adjusts the update intensity of each time step, thereby suppressing information redundancy and improving the modeling effect for long sequences while maintaining linear computational efficiency. This behavior can be expressed as follows:

$$\begin{aligned}\left\{\Delta_t^i\right\}_{i=1}^{S} &= \text{softplus}\left(\left\{W_\Delta^i \mathbf{h}_t + \mathbf{b}_\Delta^i\right\}_{i=1}^{S}\right)\\ \boldsymbol{\alpha}_t &= \text{softmax}\left(W_\alpha \mathbf{x}_t\right)\\ \Delta_t &= \sum_{i=1}^{S} \alpha_t^i \cdot \Delta_t^i\end{aligned} \tag{14}$$

where S denotes the number of scales, and $W_\Delta^i \in R^{d\times d}$ and $W_\alpha \in R^{S\times d}$ are the learnable parameters. Meanwhile, in

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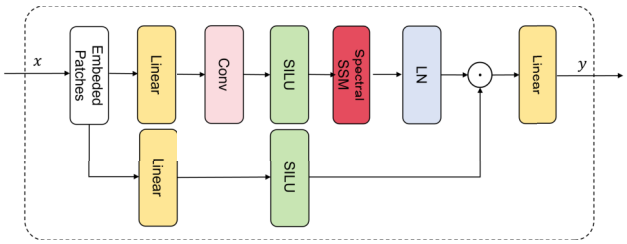


Fig. 2. Structure of mamba.

order to ensure the stability of the training process, we have implemented a stabilization mechanism

$$\overline{\mathbf{A}} = -e^{(\mathbf{A}_{\log})} \tag{15}$$

where $\mathrm{A}_{\log} \in R^{N\times N}$ represents free parameters. At the same time, we introduced a gated mechanism for selective memory retention

$$\begin{aligned} g_t &= \sigma\left(W_g x_t\right) \\ \overline{\mathbf{B}_t} &= \mathbf{B} g_t \\ \overline{\mathbf{C}_t} &= \mathbf{C} g_t \end{aligned} \tag{16}$$

where $W_g \in R^{N\times D}$, implements parameterized state transitions through gated multiplicative interactions. In the final output layer, we employed triple residual connections, enhancing feature fusion through learnable weights and local convolution, while maintaining gradient stability and improving the spectral modeling capability. The learnable weight matrix ensures the transmission of low-frequency signals, and the depthwise separable convolution captures high-frequency details, achieving full-spectrum coverage. This process is manifested as follows:

$$\begin{aligned} y_t^{\mathrm{SSM}} &= \overline{\mathbf{C}} h_t + \overline{\mathbf{D}} x_t \\ y_t &= y_t^{\mathrm{SSM}} + \beta_1 W_r x_t + \beta_2 \mathcal{F}\left(x_t\right) \end{aligned} \tag{17}$$

where $\overline{\mathbf{D}} \in R^D$ is the skip connection parameter, $W_r \in R^{D\times D}$ denotes a learnable residual weight matrix, β is a balancing hyperparameter, and $\mathcal{F}(\cdot)$ represents depthwise separable convolution.

The entire Mamba branch is shown in Fig. 2. The input fused feature $\mathbf{F}_{\mathrm{Fused}}$ is first normalized by layer normalization, which standardizes the distribution along the channel dimension to stabilize the training. Subsequently, the branches process the original feature sequence and the reversed feature sequence in parallel using Mamba blocks, achieving bidirectional spectral sequence modeling and effectively capturing the bidirectional spectral context, thereby overcoming the limitations of traditional unidirectional approaches. This process is represented as follows:

$$\begin{aligned} \mathbf{F_{M1}} &= \mathrm{Mamba}\left(\mathrm{LN}\left(\mathbf{F}_{\mathrm{Fused}}\right)\right) \\ \mathbf{F_{M2}} &= \mathrm{Mamba}\left(\mathrm{Flip}\left(\mathrm{LN}\left(\left(\mathbf{F_{M1}}\right)\right)\right)\right) \end{aligned} \tag{18}$$

where $\mathrm{LN}(\cdot)$ denotes the layer normalization process computing mean and variance along the channel dimension, and $\mathrm{Flip}(\cdot)$ represents spectral dimension reversal.

To further stabilize the feature distribution and avoid gradient disappearance, we add the bidirectional features element-wise and fuse the shallow and deep information through residual connections. Finally, it is input into a MLP for nonlinear transformation to enhance the feature expression ability, and at the same time, through skip connections, the original information is retained to prevent the loss of spatial details caused by excessive compression. The final result is the global features processed by the Mamba flow. The process is as follows:

$$\begin{aligned} \mathbf{F_{M3}} &= \mathrm{LN}\left(\mathbf{F_{M2}} + \mathbf{F_{M1}}\right) \\ \mathbf{F_M} &= \mathrm{MLP}\left(\mathbf{F_{M3}}\right) + \mathbf{F_{M3}}. \end{aligned} \tag{19}$$

2) GCN Branch: The GCN branch processes the fused features through a hierarchical geometric learning framework. To overcome the limitations imposed by the static adjacency matrix, we have devised a method to construct a dynamic adjacency matrix based on the height H and width W of the feature map, and its form is as follows:

$$\mathbf{A}_{ij} = \frac{\exp\left(-\frac{\left\|p_i - p_j\right\|_2^2}{2\sigma^2}\right)}{\sum_{k=1}^{\mathrm{HW}} \exp\left(-\frac{\left\|p_i - p_k\right\|_2^2}{2\sigma^2}\right)}. \tag{20}$$

Here, p represents the spatial coordinates of the feature point on the 2-D feature map, and σ is the bandwidth parameter of the Gaussian kernel.

After obtaining the adjacency matrix, geometric feature propagation is achieved through graph convolution operations

$$\mathbf{F_{G1}} = \mathrm{GCN}\left(\mathbf{F}_{\mathrm{Fused}}, \mathbf{A}\right). \tag{21}$$

To avoid the problem of excessive smoothing, we introduced a gated fusion mechanism, which achieves adaptive feature selection through 1×1 convolution and sigmoid activation function. This design is used to handle the issue that occurs when traditional GCN is stacked in multiple layers due to the homogeneity of features. Its form is as follows:

$$\begin{aligned} \mathbf{F_{G2}} &= \mathrm{Concat}\left(\mathbf{F_{G1}}, \mathbf{F}_{\mathrm{Fused}}\right) \\ \mathbf{F_{G3}} &= \sigma\left(\mathrm{Conv2D}_{1\times 1}\left(\mathbf{F_{G2}}\right)\right). \end{aligned} \tag{22}$$

The gating mechanism brings about two improvements: on one hand, it retains the discriminative information of the original features; on the other hand, it suppresses the interference from irrelevant neighborhood features. Finally, through residual connections, the gradient flow is guaranteed

$$\mathbf{F_G} = \mathbf{F_{G3}}\mathbf{F_{G1}} + \mathbf{F}_{\mathrm{Fused}}. \tag{23}$$

By integrating the optimized Mamba branch and the GCN branch collaboratively, a spectral–spatial combination is achieved. The final mixed features combine these complementary advantages through a direct addition method. The final processed features $\mathbf{F}_{\mathrm{HGM}}$ are obtained by

$$\mathbf{F}_{\mathrm{HGM}} = \mathbf{F_M} + \mathbf{F_G}. \tag{24}$$

D. Stage of the Classification

To enhance computational efficiency and prevent overfitting, we employ a simple linear structure for the classifier in the classification stage. Furthermore, to fully exploit the complementary information from multimodal features, we design an

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efficient multipath classification architecture that achieves optimal classification performance through an adaptive decision fusion strategy.

The initial processing stage involves feeding $\mathbf{F_M}$, $\mathbf{F_G}$, and $\mathbf{F}_{\mathrm{HGM}}$ through a dual-block CNN architecture. The initial processing block employs adaptive GAP to compress spatial dimensions to a 1×1 representation, effectively extracting per-pixel categorical features. The second block contains a FC layer that maps spectral channels to the predefined number of land cover categories C, followed by a Softmax function that converts the C-dimensional features into C-dimensional probability vectors $\mathbf{P}_{\mathrm{HGM}}$, representing the land cover category probabilities for target pixels. This process can be expressed as

$$\mathbf{P}_{\mathrm{HGM}} = \mathrm{Softmax}\left(\mathrm{FC}\left(\mathrm{Avgpool}\left(\mathbf{F}_{\mathrm{HGM}}\right)\right)\right). \tag{25}$$

Next, we apply identical processing to $\mathbf{F_M}$ and $\mathbf{F_G}$, subsequently transforming their C-dimensional features into probability vectors $\mathbf{P_M}$ and $\mathbf{P_G}$ through the Softmax function.

Finally, we perform adaptive decision fusion on the classification probabilities $\mathbf{P}_{\mathrm{HGM}}$, $\mathbf{P_M}$, and $\mathbf{P_G}$ corresponding to the encoded features $\mathbf{F}_{\mathrm{HGM}}$, $\mathbf{F_M}$, and $\mathbf{F_G}$, achieving optimal integration of these three components for improved classification results. The fused classification probability $\mathbf{P}_{\mathrm{class}}$ is computed as

$$\mathbf{P}_{\mathrm{class}} = \beta_1 \mathbf{P}_{\mathrm{HGM}} + \beta_2 \mathbf{P_M} + \beta_3 \mathbf{P_G} \tag{26}$$

where $\mathbf{P}_{\mathrm{class}}$ represents the classifier's output feature with dimensionality equal to the number of categories. The coefficients β are learnable weights that automatically balance the contributions of $\mathbf{P}_{\mathrm{HGM}}$, $\mathbf{P_M}$, and $\mathbf{P_G}$. We initialize β_1, β_2, and β_3 to 0.5 to ensure equal initial importance among the three components.

IV. Experimental Results and Analysis

This section presents systematic experimental evaluations of the proposed DAHGMN framework using three benchmark HSI and LiDAR datasets to validate its classification performance. First, we provide detailed descriptions of the experimental datasets. Second, we clearly outline the experimental configuration and quantitative evaluation metrics. Subsequently, we perform both branch-wise and componentwise ablation studies to analyze our method. Finally, we conduct extensive comparative experiments between DAHGMN and state-of-the-art methods, including quantitative comparisons, visual analysis, and accuracy evaluations under different training sample sizes, and comprehensive parameter comparisons.

A. Dataset Description

1) HSI-LiDAR Houston Dataset: The airborne dataset was acquired during a 2012 survey campaign utilizing the Compact Airborne Spectrographic Imager (CASI) sensor over the Houston urban area. This multimodal dataset integrates HSI with LiDAR-derived digital surface model (DSM) information, with spatial dimensions of 345×1905 pixels at 2.5-m ground sampling distance. The HSI data contain 144 spectral bands covering the visible to near-infrared spectrum from 0.38 to 1.05 μm. The ground reference data consist of 15 029 labeled samples across 15 urban land-use classes, exemplifying characteristic urban remote sensing challenges including spectral confusion between classes and significant within-class variability.

2) HSI-LiDAR MUUFL Dataset: Acquired during November 2010 over the University of Southern Mississippi Gulf Park Campus, this multimodal dataset integrates HSI and LiDAR-derived DSM data spanning a 325×220 pixel region. The HSI measurements were obtained using a CASI-1500 imaging spectrometer, providing 64 spectral bands from 0.38 to 1.05 μm wavelength range with 0.54×1.0 m spatial sampling. Concurrent LiDAR acquisitions were performed with an Airborne Laser Terrain Mapper (ALTM) system operating at 1064 nm wavelength, achieving 0.60×0.78 m ground resolution. The reference data contain 53 687 georeferenced samples representing 11 distinct land cover categories, capturing the spectral and structural variability characteristic of coastal urban environments and associated vegetation communities.

3) HSI-LiDAR Trento Dataset: Acquired by an airborne hyperspectral imaging systems (AISA) eagle sensor over rural areas south of Trento, Italy, this dataset features HSI and LiDAR data uniformly sampled at 1 m spatial resolution across a 600×166 pixel study area. The HSI component contains 63 spectral bands from 0.42 to 0.99 μm, while the LiDAR data comprise two distinct channels. With 30 214 georeferenced labeled samples representing six fundamental land cover classes, it characterizes spectral–spatial patterns typical of Mediterranean agro-forest ecosystems. We sampled in the standard Houston and Trento datasets, and Table I provides detailed class specifications for all three datasets.

B. Experimental Setup

All experiments in this study were implemented using the PyTorch framework and executed on an NVIDIA RTX 3090 GPU with 24 GB memory. The proposed DAHGMN was optimized using the Adam optimizer with a batch size empirically set to 64, consistent with configurations used in most comparative studies to ensure fair performance benchmarking. Training proceeded for 500 epochs, determined through observation of test accuracy convergence behavior. Performance evaluation incorporated per-class accuracy, overall accuracy (OA), average accuracy (AA), and Kappa coefficient (κ).

For fair and robust comparisons, all metrics were averaged over ten independent classification trials. When benchmarking against existing methods, we strictly followed the original implementations where parameters and training procedures were explicitly specified. For methods lacking implementation details, we applied the same training protocol as our proposed approach to ensure consistency.

C. Parameter Sensitivity Analysis

1) Patch Size: A rigorous analysis of spatial context influence was performed by evaluating model performance across multiple patch dimensions {9, 11, 13, 15, 17, 19}. Fig. 3(a) demonstrates that maximum classification accuracy was achieved with 13×13 pixel windows for both the Houston and Trento study areas, whereas the MUUFL dataset exhibited optimal results with 9×9 pixel neighborhoods.

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TABLE I
DETAILED CATEGORY INFORMATION FOR THREE DATASETS

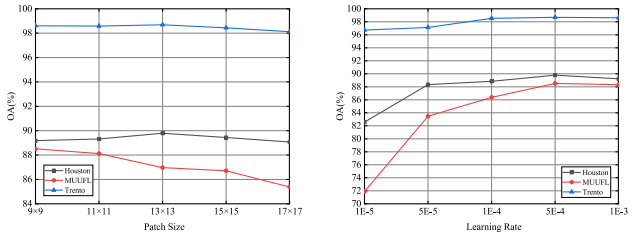
NO.	Houston			MUUFL			Trento		
	Class name	Train/All	Test	Class name	Train	Test	Class name	Train/All	Test
C1	Healthy Grass	50/198	1053	Trees	50	23196	Apple Trees	50/129	3905
C2	Stressed Grass	50/190	1064	Mostly Grass	50	4220	Buildings	50/125	2778
C3	Synthetic Grass	50/192	505	Mixed Ground Surface	50	6832	Ground	50/105	374
C4	Trees	50/188	1056	Dirt and Sand	50	1776	Woods	50/154	8969
C5	Soil	50/186	1056	Road	50	6637	Vineyard	50/184	10317
C6	Water	50/182	143	Water	50	416	Roads	50/122	3052
C7	Residential	50/196	1072	Buildings Shadow	50	2183			
C8	Commercial	50/191	1053	Buildings	50	6190			
C9	Road	50/193	1059	Sidewalk	50	1335			
C10	Highway	50/191	1036	Yellow Curb	50	133			
C11	Railway	50/181	1054	Cloth Panels	50	219			
C12	Parking Lot 1	50/192	1041						
C13	Parking Lot 2	50/184	285						
C14	Tennis Court	50/181	247						
C15	Running Track	50/187	473						
	Total	750/2832	12197	Total	550	53137	Total	300/819	29395

TABLE II
BRANCH ABLATION ANALYSIS

Case	Houston			MUUFL			Trento		
	OA(%)	AA(%)	κ×100	OA(%)	AA(%)	κ×100	OA(%)	AA(%)	κ×100
HSI	87.96	89.47	86.93	86.39	85.64	82.20	97.91	95.67	97.21
HSI+LIDAR	**89.79**	**91.16**	**88.91**	**88.61**	**88.64**	**85.11**	**98.69**	**96.77**	**98.24**

TABLE III
COMPONENTS ABLATION ANALYSIS

NO.	Component				Houston			MUUFL			Trento		
	Mamba	GCN	DA	PCA	OA(%)	AA(%)	κ×100	OA(%)	AA(%)	κ×100	OA(%)	AA(%)	κ×100
1	✗	✓	✓	✓	85.17	86.28	83.92	86.78	86.60	82.82	98.14	96.54	97.51
2	✓	✗	✓	✓	87.97	89.50	86.96	85.74	87.25	81.90	97.95	94.43	97.24
3	✓	✓	✗	✓	89.44	90.26	88.55	87.88	88.18	84.22	98.44	96.37	97.87
4	✓	✓	✓	✗	88.44	90.22	87.46	88.32	87.49	84.72	98.48	96.36	97.94
Ours	✓	✓	✓	✓	**89.79**	**91.16**	**88.91**	**88.61**	**88.64**	**85.11**	**98.69**	**96.77**	**98.24**



(a) (b)

Fig. 3. Influence of different parameters on the OA. (a) Patch size. (b) Learning rate.

2) Learning Rate: The optimization process is critically affected by the learning rate configuration, which governs both training dynamics and convergence characteristics. Experimental evaluations were conducted across a range of learning rate values {1e^{-}5, 5e^{-}5, 1e^{-}4, 5e^{-}4, 1e^{-}3, 5e^{-}3}. As evidenced in Fig. 3(b), the classification accuracy metrics reveal that optimal model performance was consistently obtained with a learning rate setting of 5e^{-}4 across all evaluated datasets.

D. Ablation Experiment

To evaluate the effectiveness of the proposed method, we conducted comprehensive ablation studies focusing on branch-level and component-level analyses, with quantitative results presented in Tables II and III, respectively. Compared with single-modal HSI processing, the multimodal data fusion approach demonstrates superior classification accuracy as shown in Table II, confirming the effectiveness of our multimodal remote sensing data fusion network architecture. The component-level ablation reveals significant performance degradation when either the Mamba module or GCN module is removed. This is quantitatively demonstrated

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TABLE IV
CLASSIFICATION ACCURACY (%) ON THE HOUSTON DATASET

NO.	Class	Performance								
		EndNet	MDL-RS	CoupledCNN	MFT	GLT	HCT	HLMamba	S2CrossMamba	DAHGMN
C1	Healthy Grass	82.91	81.77	82.94	81.67	83.10	82.24	83.10	82.34	**84.45**
C2	Stressed Grass	83.41	83.65	84.43	83.65	84.02	84.49	**85.03**	81.02	84.42
C3	Synthetic Grass	93.56	92.67	95.97	94.26	84.95	88.51	96.49	**96.53**	91.78
C4	Trees	91.86	93.18	**96.52**	89.11	94.60	96.50	93.09	95.31	94.60
C5	Soil	99.72	**100**	99.87	**100**	**100**	99.34	**100**	**100**	**100**
C6	Water	90.56	**100**	95.80	95.80	98.60	96.85	98.95	97.90	97.55
C7	Residential	90.25	76.12	90.61	94.40	**95.34**	92.49	85.59	89.51	90.25
C8	Commercial	82.34	81.58	91.17	75.59	**93.07**	88.60	87.30	75.45	82.48
C9	Road	82.50	97.34	95.95	94.88	77.42	91.84	90.58	83.30	**99.69**
C10	Highway	81.78	84.89	85.24	81.78	84.23	77.20	**90.84**	87.25	88.27
C11	Railway	65.01	65.54	**68.73**	49.23	57.82	59.17	65.37	56.71	66.67
C12	Parkinig Lot1	85.45	95.10	93.76	87.51	89.43	91.07	86.46	86.22	**97.93**
C13	Parkinig Lot2	91.05	91.58	88.77	86.67	87.37	90.88	90.88	**92.98**	90.18
C14	Tennis Court	86.23	**100**	94.87	99.19	97.57	85.83	98.28	99.60	**100**
C15	RunningTrack	98.94	98.10	94.79	91.97	**99.15**	84.57	92.02	98.41	**99.15**
	OA	85.75	87.26	89.66	85.12	86.85	86.54	87.85	85.56	**89.78**
	AA	87.04	89.43	90.63	87.05	88.44	87.19	89.60	88.17	**91.16**
	$\kappa\times100$	84.58	86.20	88.80	83.87	85.78	85.42	86.85	84.38	**88.91**

TABLE V
CLASSIFICATION ACCURACY (%) ON THE MUUFL DATASET

NO.	Class	Performance								
		EndNet	MDL-RS	CoupledCNN	MFT	GLTNet	HCT	HLMamba	S2CrossMamba	DAHGMN
C1	Trees	86.99	87.16	90.98	92.08	89.04	93.21	93.11	92.32	**93.42**
C2	Mostly Grass	76.74	67.85	81.26	70.07	83.67	**85.19**	83.10	82.27	76.04
C3	Mixed Ground Surface	64.62	76.11	68.78	70.42	**79.35**	60.77	73.53	73.71	76.05
C4	Dirt and Sand	88.35	87.81	79.90	92.79	90.48	88.01	94.90	**96.11**	91.43
C5	Roads	82.16	80.89	84.95	80.94	84.83	85.28	83.31	82.07	**87.09**
C6	Water	98.83	99.08	**100**	98.08	**100**	99.52	99.94	98.56	99.94
C7	Building Shadow	89.38	90.69	86.07	90.11	**96.93**	89.19	94.53	88.00	93.39
C8	Buildings	89.31	93.32	**96.19**	92.37	95.30	95.61	93.95	94.70	94.22
C9	Sidewalk	73.09	58.38	**83.37**	66.52	72.43	70.11	79.21	71.76	73.90
C10	Yellow Curb	91.61	65.36	87.22	83.46	84.21	80.45	**92.67**	89.47	90.79
C11	Cloth Panels	98.25	98.33	97.72	**100**	95.89	97.26	98.86	99.54	98.80
	OA	82.91	83.66	86.53	85.55	87.63	86.81	88.51	87.63	**88.61**
	AA	85.39	82.27	86.95	85.17	88.38	85.87	**89.74**	88.05	88.64
	$\kappa\times100$	78.06	78.95	82.52	81.19	83.99	82.81	85.03	83.89	**85.11**

in Table III, verifying their critical roles in spectral feature extraction and spatial context modeling, respectively. Further experimental validation shows that eliminating the DA module adversely affects classification accuracy, proving its effectiveness in LiDAR data fusion and long-range dependency modeling. Finally, PCA dimensionality reduction effectively mitigates the curse of dimensionality, with performance improvements observed across all evaluation metrics. These systematic experiments collectively demonstrate that: the Mamba module significantly enhances spectral feature representation through Spectral SSM; the GCN module effectively captures hierarchical spatial relationships in fused data; the DA module optimally integrates elevation information from LiDAR while preserving long-range contextual dependencies. All components work synergistically to achieve state-of-the-art performance.

E. Performance Comparison

To assess the technical progress and classification efficacy of the proposed DAHGMN framework, we compared DAHGMN with contemporary HSI and LiDAR joint classification approaches, encompassing feature integration methodologies such as EndNet [55] and MDL-RS [56]; decision-level fusion methods represented by CoupledCNN [31]; Transformer-based architectures including MFT [40], GLTNet [42], HCT [41]; and emerging Mamba-based approaches HLMamba [49], S2CrossMamba [50].

1) Quantitative Results and Analysis: Tables IV–VI summarize the classification performance metrics, including per-class accuracy values, OA, AA, and κ, for the proposed methods and comparison methods in the three datasets. The highest values are indicated in bold font. The experimental outcomes reveal that the proposed DAHGMN framework attains substantially improved classification results relative to existing techniques. For instance, on the Houston dataset, DAHGMN exhibits a 4.66% improvement in OA compared to MFT, while simultaneously achieving enhanced accuracy for 13 individual land cover categories. When compared to the best-performing baseline method CoupledCNN, DAHGMN shows superior OA, AA, and κ values. Notably, our method achieves

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TABLE VI
CLASSIFICATION ACCURACY (%) ON THE TRENTO DATASET

NO.	Class	Performance								
		EndNet	MDL-RS	CoupledCNN	MFT	GLTNet	HCT	HLMamba	S2CrossMamba	DAHGMN
C1	Apple Trees	99.17	98.16	99.13	96.49	98.63	96.52	**99.28**	95.49	98.17
C2	Buildings	91.32	99.57	94.22	**99.93**	99.71	**99.93**	97.01	99.73	99.89
C3	Ground	77.91	90.91	93.94	73.80	**98.93**	91.18	61.76	96.26	91.71
C4	Woods	99.75	**100**	99.85	**100**	**100**	**100**	**100**	99.28	**100**
C5	Vineyard	97.77	**100**	**100**	**100**	**100**	**100**	**100**	99.60	**100**
C6	Roads	71.49	88.83	**93.58**	77.82	89.20	81.52	89.65	89.20	90.87
	OA	95.05	98.44	98.55	96.82	98.64	97.48	98.05	97.85	**98.69**
	AA	89.65	96.24	96.79	91.30	**97.74**	94.85	91.28	96.59	96.77
	$\kappa\times100$	93.36	97.91	98.05	95.74	98.18	96.62	97.38	97.12	**98.24**

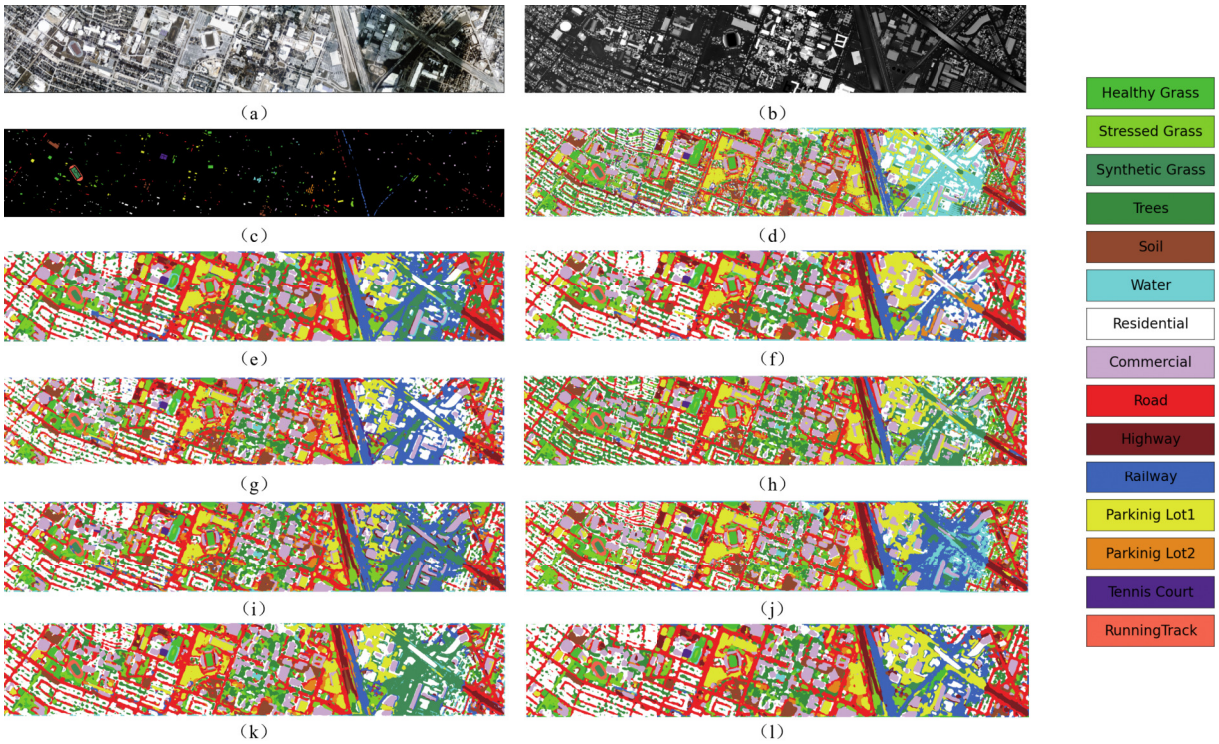


Fig. 4. Graph of classification results obtained by different methods on the Houston dataset. (a) HSI pseudo-color map, (b) LiDAR-based DSM, (c) ground truth map, (d) EndNet (85.75%), (e) MDL-RS (87.26%), (f) CoupledCNN (89.66%), (g) MFT (85.12%), (h) GLTNet (86.85%), (i) HCT (86.54%), (j) HLMamba (87.85%), (k) S2CrossMamba (85.56%), and (l) DAHGMN (89.78%).

the highest classification accuracy in six categories, while CoupledCNN excels in only two categories. This improvement stems from our multibranch feature processing and decision-level fusion strategy. The proposed method continues to perform well on the MUUFL dataset, achieving the highest accuracy for two land cover types and demonstrating significant advantages in OA and κ. On the Trento dataset, our method also achieves the best OA and κ values. These results collectively indicate that the proposed multibranch architecture effectively captures diverse spectral–spatial features through comprehensive utilization of complementary HSI and LiDAR information, as evidenced by the cross-fusion approach in MDL-RS. Comparative analysis reveals that attention-based feature integration approaches exhibit superior performance compared to conventional techniques. Specifically, architectures including HCT, GLTNet, and MFT demonstrate enhanced classification accuracy relative to standard feature-level fusion methodologies. Moreover, the incorporation of optimized decision-level fusion mechanisms contributes to additional improvements in both predictive performance and model stability. DAHGMN's consistent excellence across all evaluation metrics and datasets validates its effectiveness and advancement in multimodal remote sensing data classification.

2) Visualization Results and Analysis: The classification outcomes produced by various comparative approaches across three distinct datasets are visually depicted in Figs. 4–6. On the Houston dataset, the feature-level fusion methods exhibit the poorest visualization effects, characterized by fragmented spatial patterns and discontinuous built-up area distributions, particularly for small land cover parcels. In contrast, the classification maps generated by our DAHGMN method precisely delineate the boundaries of road segments even in

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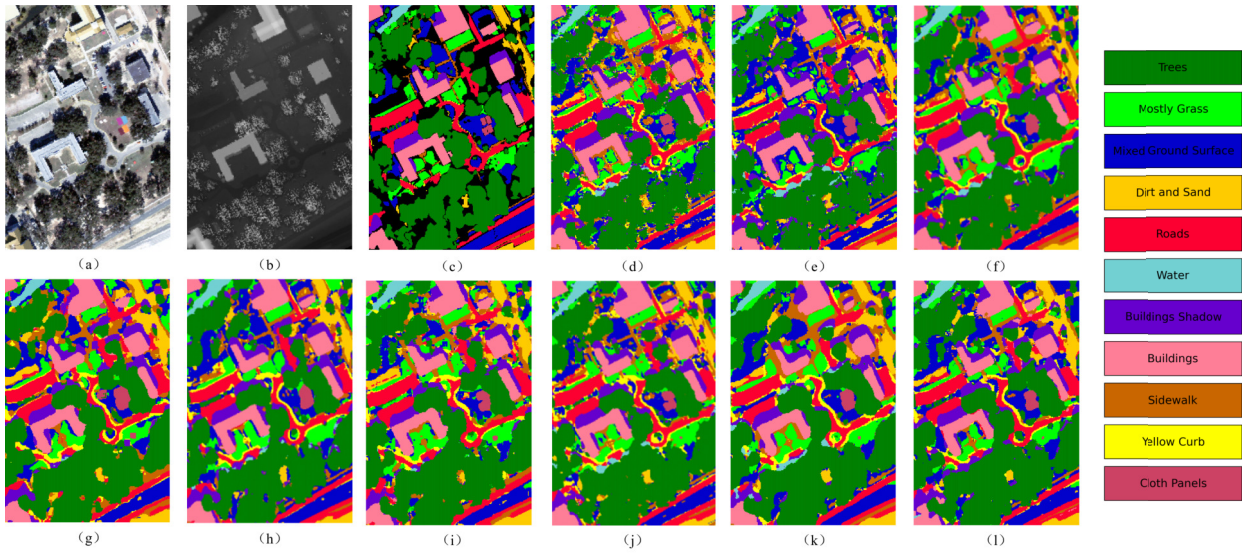


Fig. 5. Graph of classification results obtained through different methods on the MUUFL dataset. (a) HSI pseudo-color map, (b) LiDAR-based DSM, (c) ground truth map, (d) EndNet (82.91%), (e) MDL-RS (83.66%), (f) CoupledCNN (86.53%), (g) MFT (85.55%), (h) GLTNet (87.63%), (i) HCT (86.81%), (j) HLMamba (88.51%), (k) S2CrossMamba (87.63%), and (l) DAHGMN (88.61%).

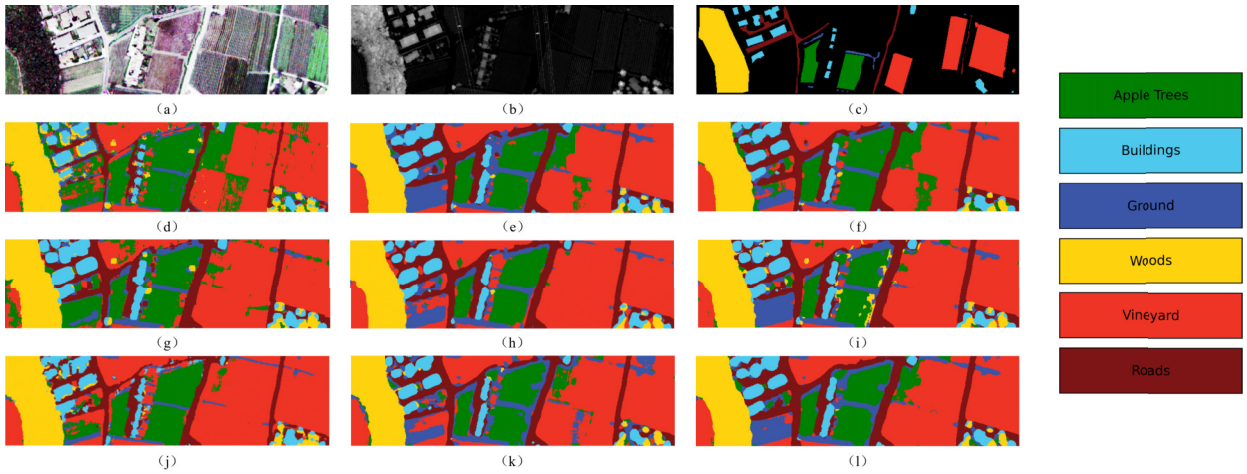


Fig. 6. Graph of classification results obtained through different methods on the Trento dataset. (a) HSI pseudo-color map, (b) LiDAR-based DSM, (c) ground truth map, (d) EndNet (95.05%), (e) MDL-RS (98.44%), (f) CoupledCNN (98.55%), (g) MFT (96.82%), (h) GLTNet (98.64%), (i) HCT (97.48%), (j) HLMamba (98.05%), (k) S2CrossMamba (97.85%), and (l) DAHGMN (98.69%).

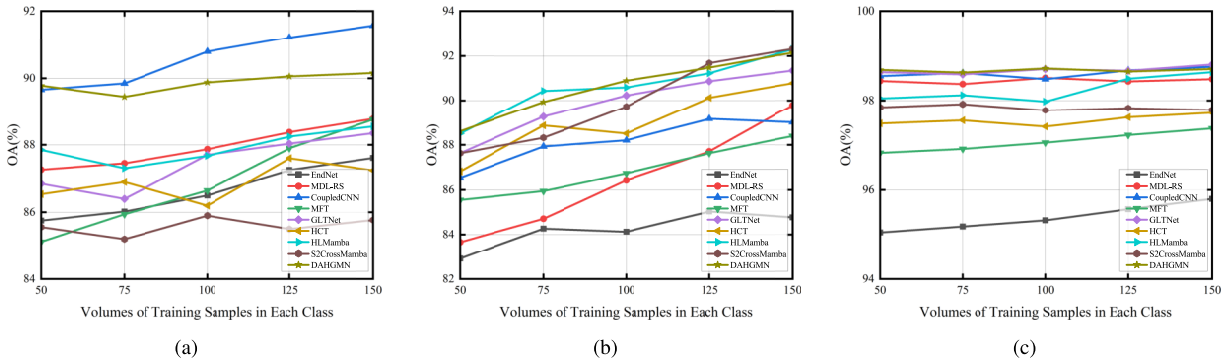


Fig. 7. Classification performance with different volumes of training samples using. (a) Houston dataset. (b) MUUFL dataset. (c) Trento dataset.

areas affected by clouds, which aligns with the quantitative results in Table IV. The superior performance confirms DAHGMN's effectiveness in fusing HSI spectral–spatial features with LiDAR elevation data. Its adaptive multimodal fusion, powered by the DA module, uniquely captures fine-grained edge details often missed by conventional methods.

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TABLE VII
COMPARISON OF MODEL COMPLEXITY OF DIFFERENT MODELS ON THE HOUSTON DATA

Method	EndNet	MDL-RS	CoupledCNN	MFT	GLTNet	HCT	HLMamba	S2CrossMamba	DAHGMN
Trainable Param (K)	90.48	120.37	102.35	313.07	560.77	955.15	221.37	379.17	271.74
Computation Cost (MFLOps)	1.11	1.66	4.57	25.30	156.54	35.96	78.43	96.76	46.56
OA (%)	85.75	87.26	89.66	85.12	86.85	86.54	87.85	85.56	89.78

Our HGM models both spatial and global spectral dependencies to extract more discriminative features, while the probabilistic decision-level fusion strategy optimally combines global and local information for final classification. For the MUUFL and Trento datasets characterized by elevated spatial resolution and limited land cover categories, DAHGMN generates classification results exhibiting exceptional agreement with reference ground truth data. On the MUUFL dataset, DAHGMN effectively utilizes global spectral dependencies to precisely distinguish mixed surfaces. In Trento's agricultural scenes, our method produces smoother results with fewer isolated pixels, particularly for the dominant vineyard class (Class 5), maintaining superior spatial continuity that accurately reflects actual planting patterns. These comprehensive visual comparisons, combined with quantitative metrics, validate DAHGMN's advancement in multimodal remote sensing data classification methodologies.

F. Comparison Under Different Training Sample Sizes

The classification accuracy is strongly influenced by the size of the training dataset. To analyze the proposed DAHGMN's performance under varying sample sizes, we conducted experiments with progressively increasing training samples (50, 75, 100, 125, and 150 samples per class), applying identical sample partitioning strategies across all methods to ensure fairness. Each experimental configuration was repeated ten times independently, with final results averaged to eliminate randomness effects. Fig. 7 illustrates the comparative performance of various classification approaches, with each method's accuracy trajectory showing consistent improvement as training data quantity increases. This empirical evidence confirms the significant performance enhancement achievable through expanded training sets in deep learning-based classification frameworks. DAHGMN not only maintains strong competitiveness with sufficient samples but also shows robust feature learning capability under limited sample conditions, demonstrating superior performance across all comparative evaluations. The results quantitatively validate DAHGMN's sample efficiency and stability in feature representation learning.

G. Computational Complexity Analysis

Table VII summarizes the architectural complexity metrics for various networks, encompassing both learnable parameters and computational requirements. The results demonstrate that our proposed model achieves remarkable efficiency regarding parameter count, ranking fifth among the nine models, being lower than the most parameter-intensive HCT while higher than the most compact EndNet. When evaluating computational efficiency using million floating-point operations (MFLOPs), the proposed method exhibits enhanced classification accuracy relative to existing techniques. This superior computational efficiency stems from the GCN module's ability to reduce computational burden for spatial feature extraction through sparse connections in local receptive fields, combined with the selective state-space mechanism in Mamba module that enables efficient modeling of long-sequence spectral data. The DA module avoids the parameter inflation problem common in traditional multibranch networks. Notably, DAHGMN achieves the best classification performance while maintaining moderate parameter scale, demonstrating excellent computation-performance trade-off characteristics.

V. CONCLUSION

This article presents a novel DL model called DAHGMN. The framework leverages a dual-stream architecture with cross-modal feature refinement, employing parallel hierarchical feature extraction branches for both HSI and LiDAR data. These branches are enhanced by a DA module that integrates spectral and spatial attention mechanisms, which effectively optimizes feature discriminability through adaptive weighting strategies. The architecture incorporates an HGM module that integrates GCN with optimized SSM-based Mamba. This integration simultaneously captures local spatial patterns through graph convolutions while modeling global spectral dependencies via state-space representations. For final classification, the network employs decision-level fusion that combines complementary features from both modalities to generate the classification results. Experimental validation demonstrates the framework's superior performance across multiple benchmark datasets. The method maintains remarkable computational efficiency while achieving these results, showing great potential for practical applications in land cover mapping and environmental monitoring. Future research will explore advanced Mamba-based fusion techniques and investigate semi-supervised learning approaches for multimodal remote sensing analysis.

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Zhenyang Xie received the B.E. degree in software engineering from Jiangxi Normal University, Nanchang, China, in 2024. He is currently pursuing the M.S. degree with the School of Information Engineering, Jiangxi University of Water Resources and Electric Power, Nanchang.

His research interests include deep learning and image classification.



Li Lv received the Ph.D. degree from Hohai University, Nanjing, China, in 2019.

She is currently a Professor with the School of Information Engineering, Jiangxi University of Water Resources and Electric Power, Nanchang, China. Her research interests include evolutionary computation, image processing, pattern recognition, and big data analysis.



Hongmin Gao (Senior Member, IEEE) received the B.S. degree in communication engineering and the Ph.D. degree in computer application technology from Hohai University, Nanjing, China, in 2006 and 2014, respectively.

He is currently a Professor with the College of Computer Science and Software Engineering, Hohai University. In the last 5 years, he was undertaking two research projects supported by the National Natural Science Foundation of China (NSFC), one research project supported by Jiangsu Natural Science Foundation, one research project supported by transformation of scientific and technological achievements in Jiangsu Province and participating the National Key Research and Development Program of China. His research interests include deep learning, information fusion, and image processing in remote sensing.

Dr. Gao's part of the research results have been applied in water resource flood disaster monitoring and operation decision supporting system of water conservancy, which won the Second Prize in Jiangxi Province Science and Technology Advancement Reward in 2018.



Shufang Xu (Member, IEEE) received the B.S. degree in applied electronic technology from Anhui Normal University, Wuhu, China, in 2002, and the master's and Ph.D. degrees in communication engineering from Nanjing University of Aeronautics and Astronautics, Nanjing, China, in 2005 and 2017, respectively.

In the last 5 years, she was participating two research projects supported by the National Key Research and Development Program of China and one research project supported by Jiangsu Province Key Research and Development Program. She is currently an Associate Professor with the College of Computer Science and Software Engineering, Hohai University, Nanjing. Her research interests include uncrewed aerial vehicle (UAV) swarm cooperation, information fusion, and image processing in remote sensing.

Dr. Xu's part of the research results have been applied in water resource flood disaster monitoring and operation decision-supporting system of water conservancy, which won China Industry-University-Research Cooperation Innovation Award in 2019 and 2021.



Haihua Xie received the B.Sc. and M.Sc. degrees in measurement and control technology and instrument from the School of Information Engineering, Jiangxi University of Water Resources and Electric Power, Nanchang, China, in 2016 and 2019, respectively, and the Ph.D. degree in control science and engineering from China University of Geoscience, Wuhan, China, in 2023.

He is currently a Lecturer with the School of Information Engineering, Jiangxi University of Water Resources and Electric Power. His current research interests include radar signal processing and machine learning.

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